

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

UCI Electricity Load Diagrams Dataset Audit Report

1. Dataset Dimensions & Metadata

- **Source File:** data/uci/LD2011_2014.txt
- **Total Customer Meters:** 370 (MT_001 to MT_370)
- **Raw Timestamps Count:** 140,256 (15-minute resolution)
- **Hourly Resampled Timestamps:** 35,065 hours
- **Date Range:** 2011-01-01 00:15:00 to 2015-01-01 00:00:00
- **Measurement Units:** Raw values are recorded in kW per 15-min interval. Values were converted to hourly aggregate energy consumption in kWh via $\sum_{i=1}^{370} \text{meter}_i / 4$.

2. Quality & Integrity Assessment

- **Missing Cells (NaN):** 0
- **Zero Consumption Cells:** 10,457,342 (20.15%)
 - *Observation:* Early 2011 records contain high zero counts because several meters were brought online progressively during 2011 and 2012.
- **Negative Values:** 0
- **Duplicate Timestamps:** 0
- **Duplicate Rows:** 0

3. Aggregate System vs Customer-Level Tradeoffs

The full 370-customer 15-minute matrix contains 51,894,720 values. For next-hour system load forecasting, aggregating the meters into a single hourly aggregate system load vector ($\text{extload}_t = \sum_{i=1}^{370} \text{load}_{i,t}$) enables high-precision grid-level demand forecasting while maintaining computational tractability.

4. Output Artifacts

- **Statistics CSV:** reports/UCI_DATASET_STATISTICS.csv
- **Processed Hourly Load:** data/processed/uci_hourly_aggregate_load.csv
- **EDA Plots:** graphs/uci/eda/